Statistical Testing and Effect Size

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The BIG question

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- give a measure of my **confidence** in the results

Statistical testing

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Statistical testing is a procedure which allows us to quantify this level of confidence

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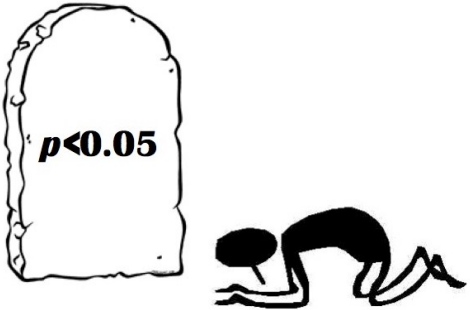
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3. Set a threshold of reasonable confidence (0.05, 0.01, ...)

p-value worship



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- We claim that a new secret drug we recently invented **significantly** reduces the cholesterol level in the population
- To prove it we get a group of 50 people, we treat them with the drug and we measure their average cholesterol level. This mean turns out to be **193**.
- Is this pilot study supporting my claim? i.e. is the difference between 193 and 200 significant?

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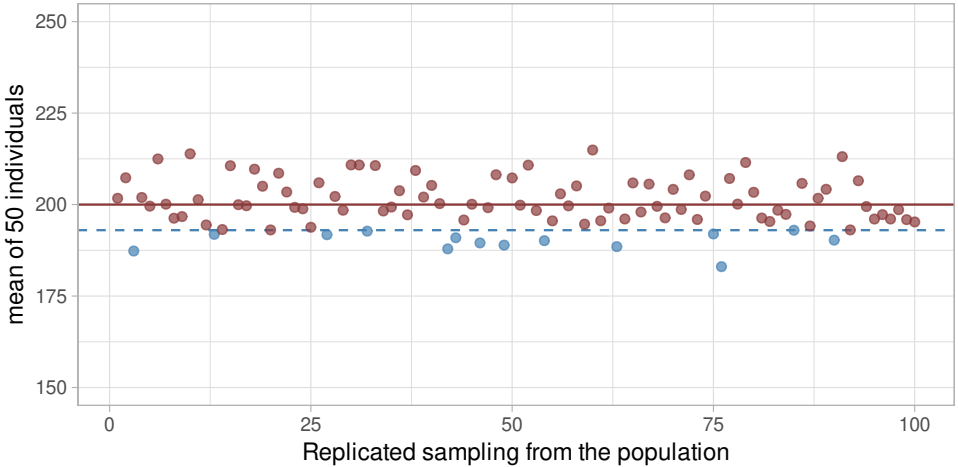
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3. Calculate the probability of obtaining less than **193** from this distribution (*p-value*)
4. Reject H0 if the p-value is lower than the selected threshold (typically 0.05)

Let's plot it!


250
225
200
175
150
mean of 50 individuals
0
25
50
75
100
Replicated sampling from the population

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I cannot reject H0 at the 0.05 level ...

More on this ...

So you see ... we need a **probability** and we also need a **threshold** (0.05)

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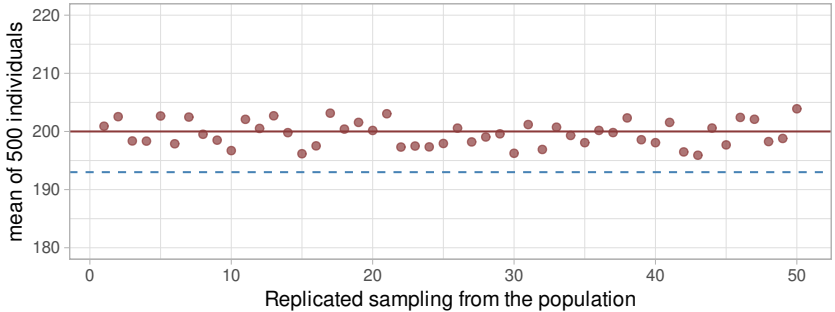
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" ... the results are not statistically significant at the 0.05 level" ...

... but they are significant at the 0.15 level!

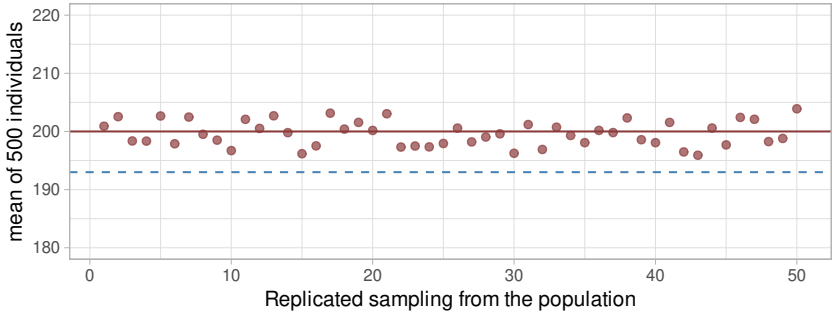
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I'm stubborn, so I'm convinced that the drug is really working. We redo the same study with a test group of **500** people and we observe once more an average level of 193 ...



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Magic! No more blue dots: now it is significant at the 0.05 level! It is extremely unlikely to get an average of 193 in a group of 500 people if the drug has no effect

Take-home messages

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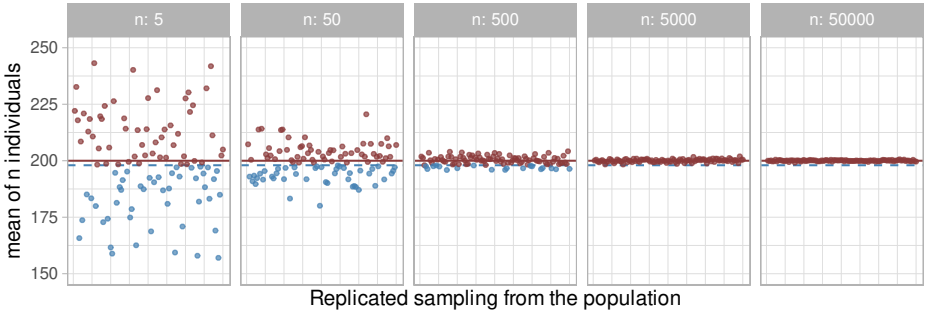
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- With more samples we can "see" smaller differences
- Is statistical significance the only thing we are looking for?

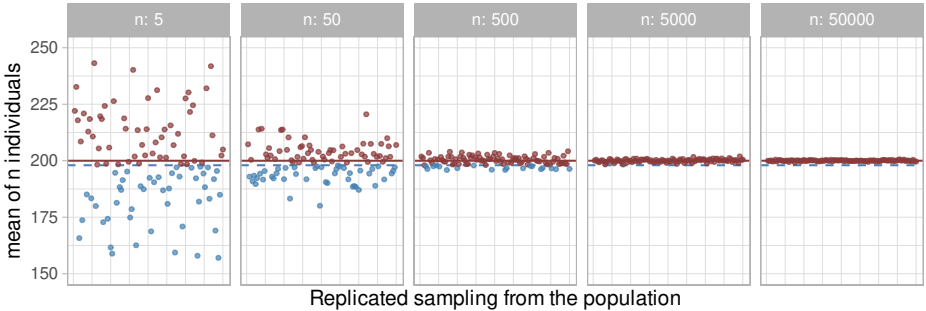
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Unfortunately it turns out that our drug is not so good ... apparently it reduces the cholesterol by only 1% (from 200 to 198)



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With 50000 people even a 1% reduction becomes "significant"

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The *p-value* alone cannot be used to judge the relevance of a research result ...

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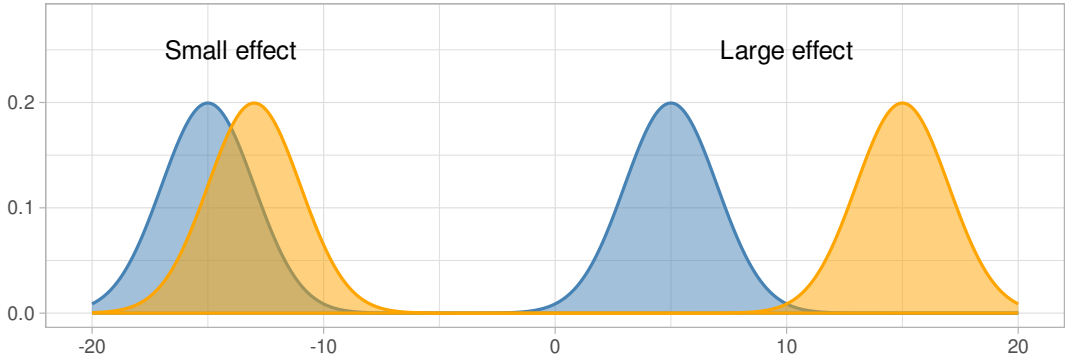
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- strong effects with low variability **will** result in low p
- the reverse is not necessarily true!
- "look, I have a low *p-value*!" is not the only thing to look for

Measuring the *effect size*



Effect size: notes

1. The difference in means is not sufficient
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Scale the log fold change by the variability → **Cohen's d on log-transformed data** (non-parametric alternative: probability of superiority)

Cohen's d

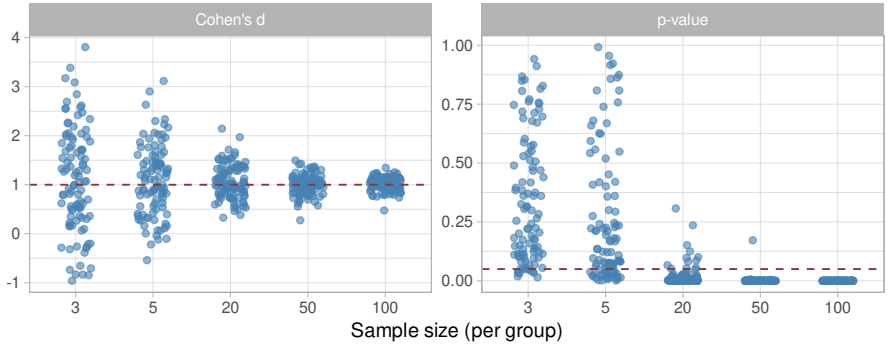
$$d = \frac{\bar{x}_1 - \bar{x}_2}{s}$$

Where

- $\bar{x}$ are the estimates of the population means
- s is the estimate of the population standard deviation (pooled)

Let's *see*

- 2 populations: $\mu_1 = 0$, $\mu_2 = 5$, $\sigma = 5$ (so the true $d = 1$)
- *t-test* to test the difference
- 100 simulated studies for each sample size



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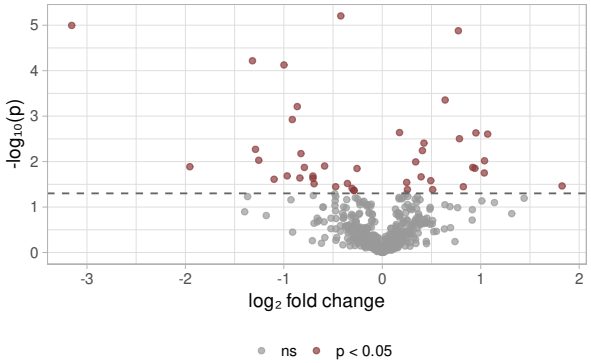
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Even the effect size does not tell me if something is relevant!

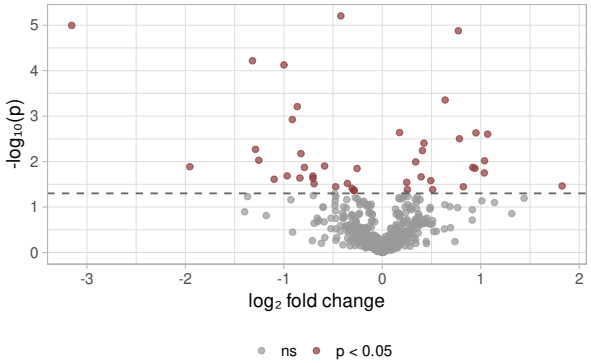
Effect size and significance: the volcano plot



Very popular in metabolomics papers:

- x-axis: $\log_2$ fold change — the *size* of the effect (?)
- y-axis: $-\log_{10}(p)$ — the *confidence* in the effect

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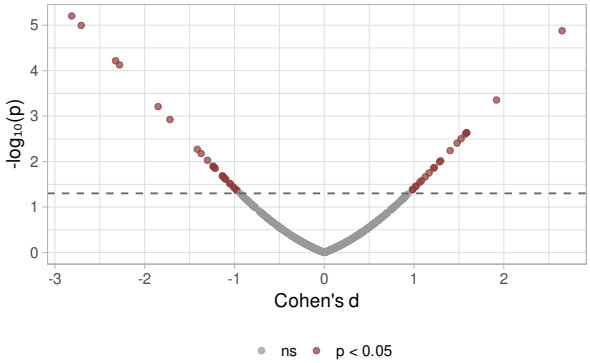


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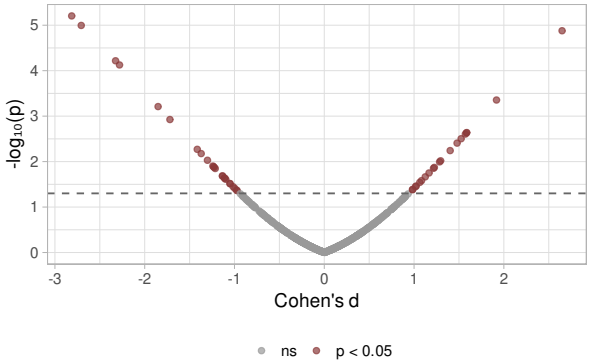
The fold change ignores the variability: the same fold change can be a huge effect for a stable metabolite and pure noise for a variable one.

Volcano plot with Cohen's d



Same data, but on the x-axis a real effect size: Cohen's d

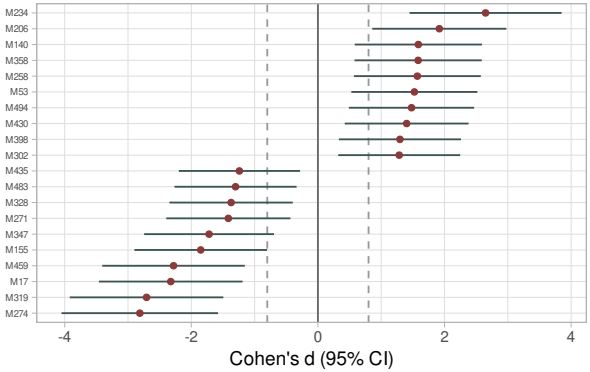
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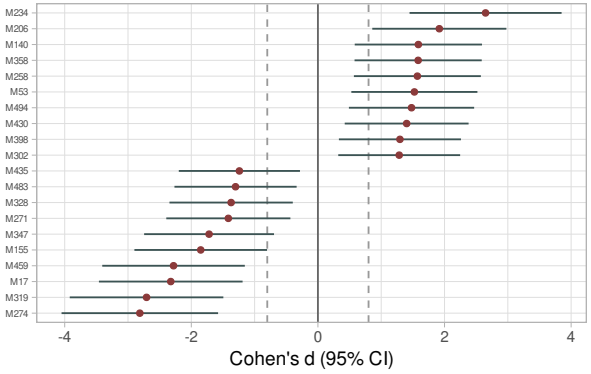
The "volcano" collapses on a curve: for a given sample size, $t = d\sqrt{n/2}$, so p-value and effect size carry the same information

Report effect sizes with their uncertainty



Cohen's d with its 95% confidence interval for the 20 metabolites with the smallest p-value

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With few samples, even "significant" effects are estimated with large uncertainty

With 500 metabolites, many of the p < 0.05 are false positives: we need to correct the p-values $\rightarrow$ **multiple testing**